Handling Missing Data in Latent Transition Analysis: A Monte Carlo Study

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Abstract

Missing data are common in longitudinal studies and can reduce the information available for statistical modeling. This is particularly consequential in latent class and mixture models because class membership is unobserved and must be inferred from patterns of observed responses; when indicators are missing, less information is available to distinguish among latent classes. These concerns extend naturally to latent transition analysis (LTA), which examines changes in latent-class membership over time. Therefore, understanding how different missing-data approaches affect recovery of LTA parameters is important, yet relatively limited guidance exists regarding their comparative performance. The present study used a Monte Carlo simulation to compare three approaches for handling missing indicator data in a two-class LTA model with binary indicators: full information maximum likelihood (FIML), multiple imputation (MI), and single stochastic imputation (SI). Simulation conditions varied sample size, number of measurement occasions, number of indicators per occasion, missing-data mechanism, and missing-data rate, with a no-missingness condition included to evaluate baseline recovery. Parameter recovery was evaluated using the magnitude of relative bias for initial latent-status probabilities, transition probabilities, and item-response probabilities. Across conditions containing missing data, FIML generally produced more accurate parameter recovery than MI and SI. Recovery improved with larger sample sizes and more indicators. Parameter-specific results further indicated that initial latent-status and item-response probabilities were more sensitive to challenging study-design and missingness conditions than transition probabilities. Overall, the findings demonstrate that both missing-data handling method and study-design characteristics influence parameter recovery in LTA and highlight the importance of adequate information in the study design when analyzing incomplete longitudinal data.

Keywords: Latent Transition Analysis, Missing Data Analysis, Full Information Maximum Likelihood, Multiple Imputation, Simulation Study

Introduction

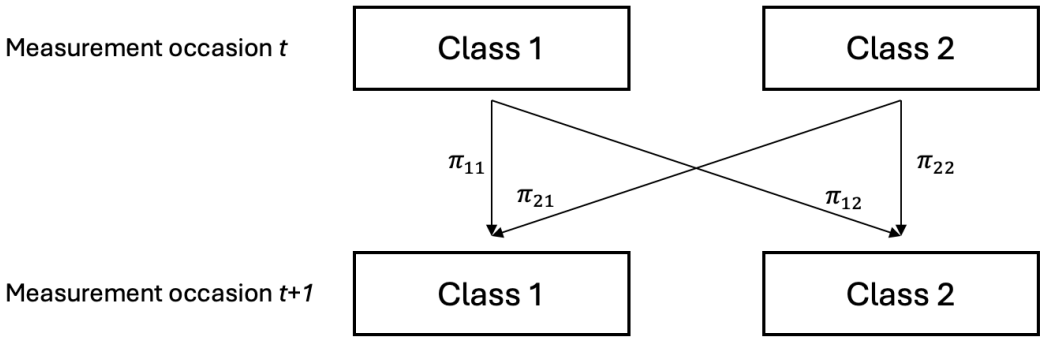
Missing data are a common challenge in longitudinal research. Because longitudinal studies require repeated observations from the same individuals over time, incomplete data may arise through participant attrition, missed assessments, item nonresponse, or other forms of nonparticipation. In a review of 100 randomized clinical trials involving longitudinal outcomes, for example, Powney et al. (2014) found that 91 of the 100 studies had missing data. Similarly, Okpara et al. (2022) reported that the proportion of missing data in longitudinal observational studies was as high as 55%. Missing observations reduce the amount of information available for estimation, which can decrease statistical power and the precision of parameter estimates (Karahalios et al., 2012). Although many methods are available for addressing missing data, they differ in their assumptions and effectiveness. Using an inappropriate method to handle missing data can introduce bias, reduce the precision of parameter estimates, and potentially lead to misleading conclusions (Karahalios et al., 2012). Consequently, appropriately handling missing data is an important consideration in longitudinal statistical modeling.

Latent Transition Analysis

One approach for examining longitudinal change in categorical outcomes is latent transition analysis (LTA). LTA is a longitudinal extension of latent class analysis in which individuals are assumed to belong to an unobserved, or latent, categorical class at each measurement occasion (Lanza & Collins, 2008; Sorgente et al., 2019; Moore et al., 2025). In LTA, these time-specific latent classes are often referred to as latent statuses. Importantly, class membership can change (e.g., an individual may belong to one class at measurement occasion *t*, but transition to another class at *t+1*). Changes in latent-status membership across occasions can then be used to characterize patterns of stability and change over time. A basic LTA model contains three primary sets of parameters: initial latent-status probabilities, transition probabilities, and item-response probabilities (Lanza & Collins, 2008; Moore et al., 2025).

Figure 1

Transition Structure of a Two-Class Latent Transition Model



Note. Arrows represent possible transitions in latent-class membership between adjacent measurement occasions. The probabilities π_{11} and π_{22} represent remaining in the same latent class, whereas π_{12} and π_{21} represent transitions between classes.

More formally, let C_t denote an individual's latent status at measurement occasion *t*, and let $\mathrm{Y_{jt}}$ denote the response to indicator *j* at occasion *t*. The initial latent-status probabilities, $\delta_{\mathrm{c}} = \mathrm{P(C_1 = c)}$, describe the probability of membership in latent status *c* at the first measurement occasion. Transition probabilities, $\pi_{\mathrm{cc'}} = \mathrm{P(C_t = c' | C_{t-1} = c)}$, describe movement from latent status *c* at one occasion to status c′at the next. For binary indicators, item-response probabilities, $\psi_{jc} = \mathrm{P\big(Y_{jt} = 1 |\, C_t = c\big)}$, characterize the relationship between the observed indicators and latent-status membership. Together, these parameters define the latent structure of the model and its evolution over time (Lanza & Collins, 2008; Moore et al., 2025).

In practice, LTA has been applied across a range of fields, including the social and behavioral sciences, education, and public health (Sorgente et al., 2019; Ryoo et al., 2018; Scorza et al., 2015). For example, Ryoo et al. (2018) used LTA to identify distinct patterns of students' concerns about bullying and to examine how students transitioned among these patterns over three semesters. Identifying these distinct patterns and how they changed over time can help inform more targeted bullying-prevention efforts for students with different patterns of concern (Ryoo et al., 2018).

Missing data are particularly consequential in latent class and latent transition models because latent-status membership is not directly observed but inferred from patterns of responses to observed indicators. When indicator responses are missing, less information is available to distinguish among latent statuses, introducing an additional source of uncertainty beyond the fact that latent-status membership itself is unobserved. Research on latent class regression has similarly identified nonresponse on manifest indicators as an important source of missing information and has examined likelihood- and imputation-based approaches for addressing it

(Harel et al., 2013). In LTA, incomplete indicator patterns may affect estimates of latent-status prevalence, item-response probabilities, and transitions between statuses over time. This issue can become particularly consequential when the missingness process is related to latent-status membership. For example, Sterba (2016) demonstrated that ignoring latent-state-dependent nonignorable missingness could distort estimates of latent-state membership; in her empirical illustration, a conventional LTA overestimated the proportion of individuals ending in one latent state relative to a model that explicitly represented the missingness process. Together, these findings highlight the importance of appropriately accounting for missing observations when estimating latent categorical structures and their transitions over time.

Importantly, the appropriateness of a method for handling missing data depends in part on the mechanism generating the missingness. Missing-data mechanisms are commonly distinguished as missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR; Little & Rubin, 2002). Under MCAR, the probability that an observation is missing is unrelated to both observed and unobserved information. Under MAR, missingness may depend on observed information but, after conditioning on that information, does not additionally depend on the missing value itself. In contrast, under MNAR, the probability of missingness depends on unobserved information or on the missing value itself. For example, suppose a longitudinal study measures a binary indicator across several occasions. If missingness at later occasions occurs independently of all observed and unobserved responses, the data are MCAR. If missingness depends on the individual's fully observed response at the first occasion, the data may be MAR. However, if the probability of missingness depends on the individual's unobserved response at the occasion being missed, the data are MNAR. These distinctions are important because methods that yield valid estimates under MCAR or MAR may not perform appropriately when the missing-data process is MNAR. The present study focuses on MCAR and MAR, two mechanisms under which likelihood- and imputation-based approaches can be applied without explicitly modeling a nonignorable missingness process.

Methods for Addressing Missing Data

The present study considers three approaches to handling missing data: full information maximum likelihood (FIML), multiple imputation (MI), and single stochastic imputation (SI). FIML estimates model parameters directly from the observed data rather than replacing missing observations with imputed values. Let $Y_{i,obs}$ and $Y_{i,mis}$ denote the observed and missing portions of the data for individual *i*, respectively. For discrete responses, the observed-data likelihood can be expressed as

$$L(\theta; Y_{obs}) = \prod_{i=1}^{n} \sum_{Y_{i,mis}} f\left(Y_{i,obs}, Y_{i,mis}; \theta\right)$$

where θ denotes the model parameters and the summation is taken over the possible values of the missing responses. FIML estimates θ by maximizing this observed-data likelihood, allowing individuals with partially observed data to contribute all available information to estimation (Enders & Bandalos, 2001). Under appropriate assumptions regarding the model and missing-data mechanism, FIML can provide consistent and efficient parameter estimates without explicitly imputing the missing observations.

MI approaches missing data by generating multiple plausible values for each missing observation rather than estimating model parameters directly from the incomplete data. Specifically, an imputation model is used to generate *M* completed datasets, each reflecting uncertainty about the missing values. The substantive model is then fitted separately to each completed dataset, producing estimates $\widehat{\mathrm{Q}}_1, \ldots, \widehat{\mathrm{Q}}_M$. The corresponding parameter estimates are combined by taking their arithmetic mean across the *M* imputed datasets, with the pooled point estimate given by

$$\bar{\mathrm{Q}} = \frac{1}{\mathrm{M}} \sum_{m=1}^{M} \widehat{\mathrm{Q}}_m$$

In standard applications of MI, uncertainty associated with the imputation process can also be incorporated using Rubin's pooling rules (Rubin, 1987). Let U_m denote the estimated sampling variance of $\widehat{\mathrm{Q}}_m$ within imputed dataset *m*. The average within-imputation variance is

$$\bar{\mathrm{U}} = \left(\frac{1}{\mathrm{M}}\right) \sum_{m=1}^{M} \mathrm{U_m}$$

and the between-imputation variance is

$$B = \frac{1}{M-1} \sum_{m=1}^{M} \left(\widehat{\mathrm{Q}}_\mathrm{m} - \bar{\mathrm{Q}}\right)^2$$

These components are combined to obtain the total variance,

$$T = \bar{\mathrm{U}} + \left(1 + \frac{1}{M}\right) B$$

Thus, standard MI inference can reflect both sampling variability within completed datasets and variability across imputations. The present study focuses on recovery of LTA point estimates rather than standard errors or hypothesis testing; as such, the analyses use only the pooled point estimate $\bar{\mathrm{Q}}$. Of note, the validity of MI also depends on the appropriate specification of the imputation model, including preservation of relationships among variables that are important for the substantive analysis.

Single imputation (SI) provides a simpler approach to MI in which each missing value is replaced once to create a single completed dataset. Several forms of single imputation have been used, including mean-based imputation and donor-based approaches in which missing values are replaced using observations from similar cases (de Waal et al., 2011). A stochastic alternative predicts each missing value using an imputation model and then draws a single plausible value from the resulting predictive distribution. In the present study, single stochastic imputation was implemented using the same logistic-regression imputation framework as MI. For each missing binary indicator response, the remaining observed indicator variables were used as predictors in a logistic regression model, producing an estimated probability of a response of 1. A single response was then drawn from a Bernoulli distribution using that probability, resulting in one completed dataset. This procedure is analogous to the MI approach used in the present study, except that SI generates only one completed dataset rather than multiple completed datasets. Consequently, SI does not average LTA parameter estimates across multiple plausible imputations, so the uncertainty associated with a particular stochastic imputation is retained in the resulting point estimate.

Previous studies have found that likelihood-based approaches and MI can perform similarly when their respective models are correctly specified. For example, Harel et al. (2013) reported comparable results for maximum likelihood and MI in latent class regression, while Lee and Shi (2021) found similar performance in correctly specified structural equation models. However, the relative performance of these methods may depend on the modeling context, and findings from other latent-variable models may not generalize directly to LTA.

Despite the extensive literature on missing data more broadly (Schafer & Graham, 2002; Lee & Shi, 2021), relatively limited guidance is available regarding how commonly used missing-data methods perform specifically in LTA. Existing research has examined missingness in structural equation models (Lee & Shi, 2021), latent class models (Harel et al., 2013), and specialized LTA settings involving nonignorable missingness (Sterba, 2016), but less is known about the recovery of basic LTA parameters under common missing completely at random (MCAR) and missing at random (MAR) conditions. In particular, the performance of missing-data methods may vary across different study-design and missingness conditions, limiting the guidance available to researchers applying LTA in practice.

Current Study

LTA provides a useful framework for examining changes in latent categorical statuses over time, but missing data can reduce the information available for estimating latent-status membership and transition patterns. Although several approaches are available for handling missing data, existing research provides relatively limited guidance regarding their performance specifically in LTA, particularly across different study-design and missingness conditions. The current study

addresses this gap through a Monte Carlo simulation comparing full information maximum likelihood (FIML), multiple imputation (MI), and single stochastic imputation (SI) for handling missing indicator data in LTA. A two-class LTA model with binary indicators is examined across conditions varying in sample size, number of measurement occasions, number of indicators, missing-data mechanism, and missing-data rate. The study evaluates how these conditions affect recovery of initial latent-status, transition, and item-response parameters and whether the performance of FIML, MI, and SI differs across conditions. The findings may provide practical guidance for applied researchers using LTA on how best to handle missing data.

Methods

Simulation Design and Data Generation

Simulation Design

A Monte Carlo simulation study was conducted to evaluate the effects of missing data and missing-data handling methods on parameter recovery in latent transition analysis (LTA). Data were generated from a two-class LTA model with binary indicators. The simulation varied sample size (n = 100, 200, 500, 1000), number of measurement occasions (T = 3, 5, 10), and number of indicators per occasion (p = 1, 5, 10). Missing-data conditions varied by missing-data mechanism and missing-data rate. For each combination of simulation conditions, 100 independent replications were generated.

Latent-Status Generation

For each replication, latent-status membership was generated according to a first-order Markov process. At the first measurement occasion, individuals were assigned to the two latent statuses with equal probabilities, such that $\mathrm{P}(\mathrm{C}_1 = 1) = \mathrm{P}(\mathrm{C}_1 = 2) = 0.50$. At subsequent measurement occasions, latent status was generated conditional on status membership at the preceding occasion. The transition-probability matrix was

$$P = \begin{pmatrix} 0.80 & 0.20 \\ 0.25 & 0.75 \end{pmatrix}$$

Thus, individuals in Status 1 had a probability of 0.80 of remaining in Status 1 and 0.20 of transitioning to Status 2, whereas individuals in Status 2 had a probability of 0.75 of remaining in Status 2 and 0.25 of transitioning to Status 1. The transition probabilities were assumed to be time-homogeneous, such that the same transition matrix was used between all measurement occasions.

Indicator Generation

At each measurement occasion, *p* binary indicators were generated conditional on latent-status membership. For all indicators, the probability of a response of 1 was 0.80 for individuals in

Status 1 and 0.30 for individuals in Status 2. Thus, the two latent statuses represented groups with relatively high and low probabilities of a response of 1, respectively. Item-response probabilities were held constant across measurement occasions, and responses were generated from Bernoulli distributions conditional on latent status.

Missing-Data Generation

MCAR Missingness

After each complete dataset was generated, missingness was imposed on the observed indicator variables. Three missing-data conditions were considered: no missingness, missing completely at random (MCAR), and missing at random (MAR). Under the MCAR and MAR conditions, target missing-data rates of 10%, 20%, and 40% were examined. Indicators at the first measurement occasion were kept fully observed in all conditions. Under MCAR, each indicator response from the second measurement occasion onward was independently designated as missing with probability equal to the specified missing-data rate.

MAR Missingness

Under the MAR condition, missingness at measurement occasions $t \geq 2$ was generated as a function of the corresponding indicator value at the first measurement occasion. Specifically, for indicator *j*, the probability that the response at occasion *t* was missing depended on the fully observed baseline response $\mathrm{Y_{ij1}}$. For each indicator and measurement occasion, the intercept of the missingness model was calibrated so that the average probability of missingness equaled the target missing-data rate.

Model Estimation

FIML Analysis

For the FIML analysis, models were estimated using the LMest R package. The true latent-status variables used during data generation were removed prior to model estimation. A two-status latent Markov model was fitted to each dataset using a time-homogeneous specification. Five random starting values were used for each model fit. Initial latent-status probabilities, transition probabilities, and item-response probabilities were then extracted for subsequent evaluation of parameter recovery.

Multiple Imputation

For the MI analysis, missing indicator responses were imputed using multivariate imputation by chained equations with the mice R package. All binary indicator variables were treated as factors, and logistic regression was specified as the conditional imputation method for each variable. The predictor matrix included the remaining indicator variables as predictors in the

conditional imputation models. Twenty imputed datasets were generated for each simulated dataset using five iterations of the chained-equations algorithm.

Each of the 20 completed datasets was subsequently fitted using the same two-status LTA specification used in the FIML analysis. To address label switching across imputations, latent-status labels were aligned according to the average item-response probability within each status, with Status 1 defined as the status having the higher average probability of a response of 1. The initial-status, transition, and item-response probability estimates were then combined by taking the arithmetic mean of the corresponding estimates across successfully fitted imputed datasets.

Single Stochastic Imputation

For the SI analysis, missing indicator responses were imputed using the same logistic-regression imputation framework as in the MI analysis, but only one completed dataset was generated for each simulated dataset, $M = 1$. Binary indicator variables were treated as factors, and the remaining indicator variables were used as predictors in the conditional imputation models. The resulting completed dataset was fitted using the same two-status LTA specification used in the FIML and MI analyses. Latent-status labels were aligned according to the average item-response probability within each status. Because only one imputed dataset was generated, parameter estimates were obtained directly from the single fitted LTA model and no averaging across imputations was performed.

Performance Evaluation

Relative Bias and Absolute Relative Bias

Parameter recovery was evaluated using relative bias. For parameter θ, relative bias in replication *r* was calculated as

$$\mathrm{RB_r}\left(\hat{\theta}\right) = \left(\widehat{\theta_r} - \theta\right)/\theta$$

where $\widehat{\theta_r}$ denotes the estimated parameter in replication *r* and θ denotes its true population value. Relative bias was then averaged across the 100 replications within each combination of sample size, number of measurement occasions, number of indicators, missing-data mechanism, missing-data rate, and analysis method. Values closer to zero indicate more accurate parameter recovery, whereas positive and negative values indicate overestimation and underestimation, respectively.

For the primary results, the magnitude of condition-specific bias was summarized using absolute relative bias. Specifically, after relative bias was averaged across the 100 replications within each simulation condition, the absolute value of the mean relative bias was taken:

$$\mathrm{ARB} = \overline{|\mathrm{RB}|}$$

Thus, values closer to zero indicate more accurate parameter recovery, whereas larger values indicate greater bias regardless of the direction of estimation error.

Results

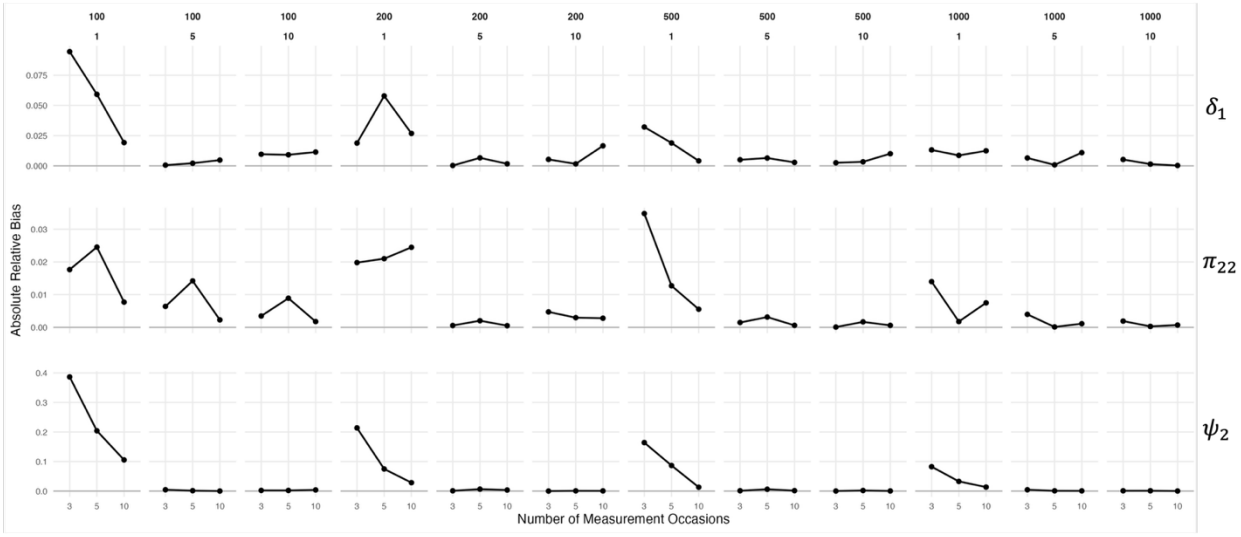
Parameter recovery was evaluated using absolute relative bias across the simulation conditions. Because several parameters are complementary in the two-class model, the main text presents results for initial, transition, and item-response probabilities, with results for the remaining parameters provided in the Supplementary Materials. Results are first presented for the no-missingness condition to characterize baseline parameter recovery, followed by a comparison of MCAR and MAR and then more detailed comparisons of FIML, MI, and SI under MAR.

No-Missingness Conditions

Under no missingness, absolute relative bias was small when five or ten indicators were available, indicating accurate baseline parameter recovery across most sample sizes and numbers of measurement occasions (Figure 1). Larger biases occurred primarily when only one indicator was available. The magnitude of bias differed substantially across parameter types. This pattern was most pronounced for the item-response probability ψ_2, which exhibited greater bias than the representative initial probability δ_1 and transition probabilities π_{22} under several *p = 1* conditions. For this parameter, bias decreased as the number of measurement occasions increased, particularly at smaller sample sizes. The initial probability also showed increased bias in some *p = 1* conditions, whereas the transition probability remained comparatively well recovered across most conditions. Increasing the number of indicators from one to five produced a marked reduction in bias for all three parameter types, while differences between five and ten indicators were small. Overall, the no-missingness results indicate that designs with only one indicator can yield noticeable estimation error even before missingness is introduced.

Figure 2

Absolute Relative Bias Under No-Missingness Conditions



Note. Absolute relative bias is shown for the initial latent-status probability δ_1, transition probability π_{22}, and item-response probability ψ_2 across sample sizes *n = 100, 200, 500, 1000*, numbers of indicators per occasion *p = 1, 5, 10*, and numbers of measurement occasions *T = 3, 5, 10*. Y-axis scales differ across parameter rows and should be interpreted separately. The horizontal reference line indicates zero absolute relative bias. Results for the remaining parameters are provided in the Supplementary Materials.

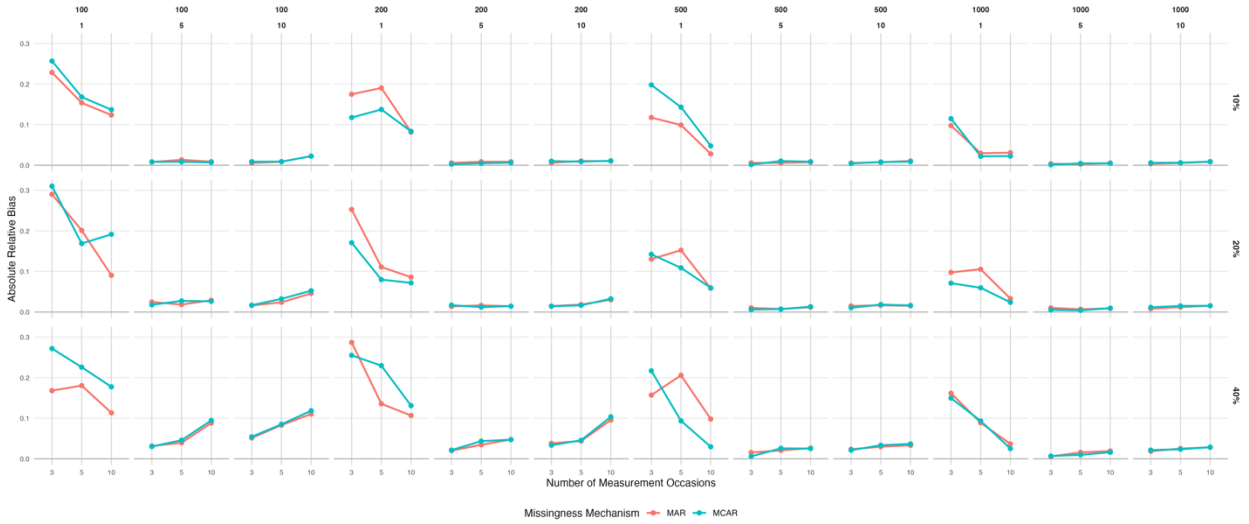
Missing-Data Conditions

Comparison of MAR and MCAR

Across the model parameters, absolute relative bias showed similar patterns under MAR and MCAR across the simulation conditions. Figure 3 illustrates this pattern for the item-response ψ_2, with results for the remaining parameters provided in the Supplementary Materials. For the item-response ψ_2, bias under both mechanisms was greatest when only one indicator was available, particularly at smaller sample sizes and higher missing-data rates. Within many of these p = 1 conditions, bias decreased as the number of measurement occasions increased. In contrast, when five or ten indicators were available, bias was substantially smaller and often remained close to zero across measurement occasions. Parameter recovery also generally improved with increasing sample size, although some condition-specific variation remained, particularly under 40% missingness. There was no consistent pattern indicating systematically greater bias under either MAR or MCAR across the design conditions examined. Accordingly, subsequent parameter-specific results focus on MAR for conciseness, with corresponding MCAR results provided in the Supplementary Materials.

Figure 3

Absolute Relative Bias for the Item-Response Probability ψ_2 Under MAR and MCAR



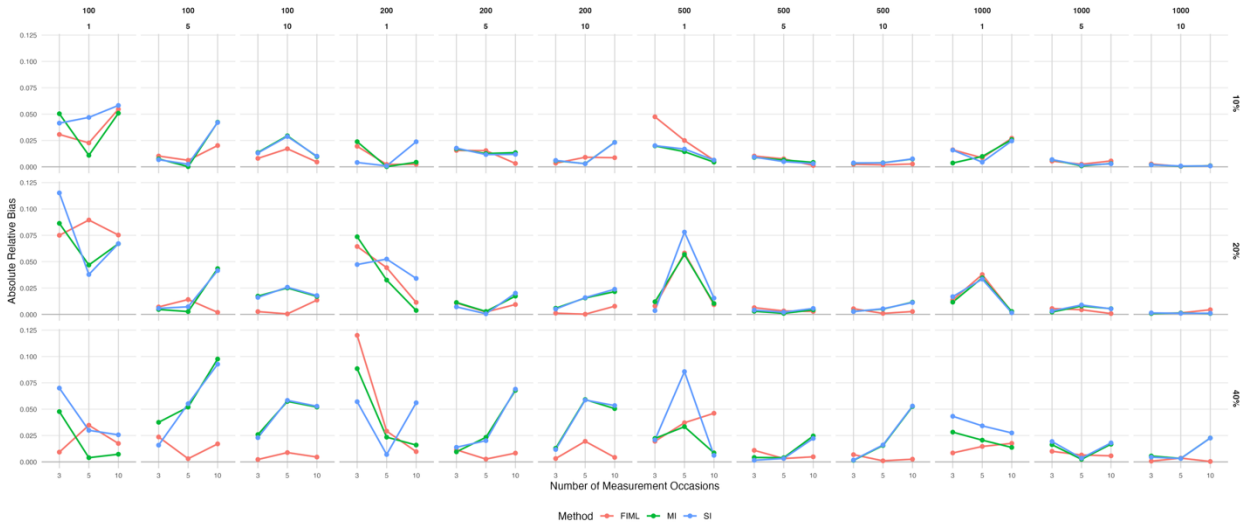
Note. Absolute relative bias for the item-response probability ψ_2 is shown under missing at random (MAR) and missing completely at random (MCAR). Values are averaged across the three missing-data methods (FIML, MI, and SI). Results are shown across sample sizes *n = 100, 200, 500, 1000*, missing-data rates of 10%, 20%, and 40%, numbers of indicators per occasion *p = 1, 5, 10*, and numbers of measurement occasions *T = 3, 5, 10*. Results for MAR–MCAR comparisons of the remaining parameters are provided in the Supplementary Materials.

Parameter Recovery Under MAR

Absolute relative bias for the initial latent-status probability δ_1 was small under lower missing-data rates (Figure 3). Larger biases occurred most often when only one indicator was available, particularly at smaller sample sizes and higher missingness rates. Increasing the number of indicators from one to five substantially reduced bias across most conditions, while additional improvement from five to ten indicators was smaller. Increasing sample size also tended to improve recovery, although some variation across numbers of measurement occasions remained. Differences among the three missing-data methods were most visible under the more challenging conditions. FIML frequently produced the smallest absolute bias, especially when missingness was high, whereas MI and SI often showed similar levels of error. When five or ten indicators were available and sample sizes were moderate to large, all three methods recovered the initial probability with relatively little bias.

Figure 4

Absolute Relative Bias for the Initial Latent-Status Probability δ_1 Under MAR

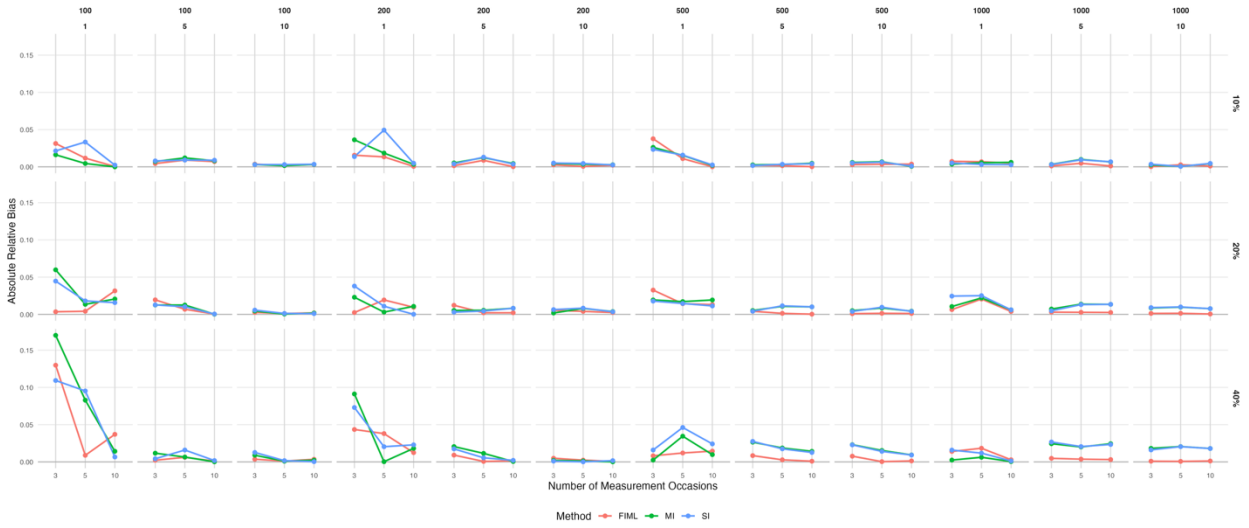


Note. Absolute relative bias is shown for full information maximum likelihood (FIML), multiple imputation (MI), and single stochastic imputation (SI) under missing at random (MAR). Results are shown across sample sizes *n = 100, 200, 500, 1000*, missing-data rates of 10%, 20%, and 40%, numbers of indicators per occasion *p = 1, 5, 10*, and numbers of measurement occasions *T = 3, 5, 10*. Lower values indicate more accurate parameter recovery.

Because the transition probabilities showed similar recovery patterns, π_{22} is presented as a representative transition parameter in the main text, with results for the remaining transition probabilities provided in the Supplementary Materials. The transition probability π_{22} as recovered with relatively small absolute bias across the MAR conditions (Figure 4). Bias remained close to zero for most conditions with five or ten indicators, even as the missing-data rate increased. Larger deviations were concentrated in designs with only one indicator, particularly when sample sizes were small and 40% of observations were missing. Under these more challenging conditions, increasing the number of measurement occasions often reduced bias, although the pattern was not completely uniform across methods and sample sizes. FIML showed the smallest bias in the conditions where the three methods differed most, whereas MI and SI tended to produce similar or slightly larger errors. Increasing sample size further reduced the differences among methods, and recovery was strong at *n = 500* or *1000*. Overall, recovery of transition probability π_{22} was relatively robust across most of the conditions examined.

Figure 5

Absolute Relative Bias for the Transition Probability π_{22} Under MAR

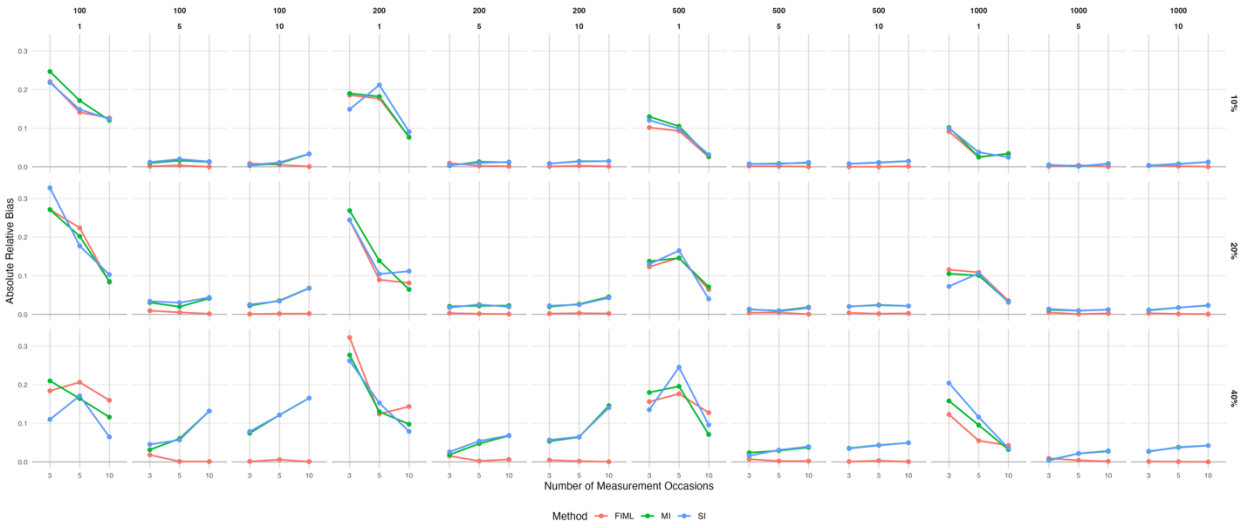


Note. Absolute relative bias is shown for full information maximum likelihood (FIML), multiple imputation (MI), and single stochastic imputation (SI) under missing at random (MAR) across sample size, missing-data rate, number of indicators per occasion, and number of measurement occasions. Lower values indicate more accurate parameter recovery. Results for the complementary transition probability are provided in the Supplementary Materials.

Absolute relative bias for the item-response probability ψ_2 was larger and more sensitive to study-design conditions than bias for the initial and transition probabilities (Figure 5). The largest biases occurred when only one indicator was available, particularly at smaller sample sizes and higher missing-data rates. Under the most challenging conditions, absolute relative bias exceeded 30%. With many of these *p = 1* conditions, increasing the number of measurement occasions reduced bias, although some irregularity remained under 40% missingness. Increasing the number of indicators from one to five produced a substantial improvement in recovery across all three methods. With five or ten indicators, FIML yielded very small bias, whereas MI and SI retained somewhat larger errors in several conditions, especially when missingness was high. Increasing sample size also improved recovery, with differences among methods becoming smaller at *n = 500 and* 1000. Overall, item-response probability ψ_2 was the most difficult of the representative parameters to recover accurately, particularly in sparse designs with substantial missingness.

Figure 6

Absolute Relative Bias for the Item-Response Probability ψ_2 Under MAR



Note. Absolute relative bias is shown for full information maximum likelihood (FIML), multiple imputation (MI), and single stochastic imputation (SI) under missing at random (MAR) across sample size, missing-data rate, number of indicators per occasion, and number of measurement occasions. Lower values indicate more accurate parameter recovery. Results for the remaining item-response probability are provided in the Supplementary Materials.

Discussion

Incomplete longitudinal data can pose challenges for latent transition analysis because both latent-status membership and transitions over time must be inferred from incomplete observed responses. Although several approaches are available for handling missing indicator data, relatively little research has directly compared their performance for parameter recovery in LTA across different study-design and missingness conditions. The present study therefore used a Monte Carlo simulation to compare FIML, MI, and SI across variations in sample size, number of measurement occasions, number of indicators, missing-data mechanism, and missing-data rate in a two-class LTA model. Overall, FIML provided the most accurate parameter recovery across the conditions examined, whereas MI and SI showed similar levels of bias. Thus, among the three approaches evaluated, FIML appears to be a strong option for handling incomplete indicator.

Study-design characteristics had substantial effects on parameter recovery. Larger sample sizes reduced absolute relative bias, particularly in conditions with fewer indicators and higher missing-data rates. Increasing the number of measurement occasions also reduced bias in many conditions, although this pattern was not uniform across parameters or methods. The number of indicators had the strongest effect on parameter recovery. Models with only one indicator per occasion showed noticeably greater bias, whereas increasing the number of indicators from one to five substantially improved recovery. Additional gains from five to ten indicators were

smaller. Parameter recovery also deteriorated as the missing-data rate increased. For applied researchers, these findings highlight the importance of using multiple indicators to adequately represent the latent classes, while also suggesting diminishing returns from adding indicators beyond five.

Parameters were not equally sensitive to the study-design and missingness conditions. Transition probabilities were recovered with relatively small bias across most conditions, whereas larger deviations were observed for the initial latent-status and item-response probabilities. In particular, the item-response probability exhibited the largest biases among the representative parameters across several challenging conditions. These differences suggest that the consequences of missing data may depend on which component of the LTA model is of substantive interest. Researchers whose conclusions depend heavily on initial latent-status or item-response parameters may therefore need to exercise greater caution when interpreting estimates under high missingness or designs with very few indicators.

Although FIML produced the smallest bias across most conditions, differences associated with study-design characteristics—particularly the number of indicators, sample size, and missing-data rate—were often more pronounced than differences among the three missing-data methods. MI and SI showed broadly similar levels of recovery, whereas FIML was especially advantageous under several of the more challenging conditions. Recovery patterns were also similar under MAR and MCAR, with no consistent indication that one mechanism produced systematically greater bias across the conditions examined.

Several limitations of the present study suggest directions for future research. The simulation focused on a relatively simple two-class LTA model with binary indicators, and the findings may not generalize directly to models with more latent classes, more complex transition structures, or different types of observed indicators. The study also considered MCAR and MAR mechanisms; future work could examine nonignorable missingness mechanisms, including MNAR settings, in which the probability of missingness depends on unobserved values or latent status. The MAR mechanism examined in the present study was also relatively simple, with missingness at later occasions depending only on fully observed baseline indicator responses. This simplicity may partly explain the limited differences observed between MAR and MCAR. Future studies could consider more complex MAR conditions in which missingness depends on additional observed indicators or covariates. In addition, the present evaluation focused primarily on relative bias and its absolute magnitude as measures of parameter recovery. Future studies could compare the methods using additional criteria such as confidence-interval coverage, hypothesis testing, and class-assignment accuracy.

References

de Waal, T., Pannekoek, J., & Scholtus, S. (2011). Handbook of statistical data editing and imputation. John Wiley & Sons.

Enders, C. K., & Bandalos, D. L. (2001). The relative performance of full information maximum likelihood estimation for missing data in structural equation models. Structural Equation Modeling: A Multidisciplinary Journal, 8(3), 430–457.

Harel, O., Chung, H., & Miglioretti, D. (2013). Latent class regression: Inference and estimation with two-stage multiple imputation. Biometrical Journal, 55(4), 541–553.

Karahalios, A., Baglietto, L., Carlin, J. B., English, D. R., & Simpson, J. A. (2012). A review of the reporting and handling of missing data in cohort studies with repeated assessment of exposure measures. BMC Medical Research Methodology, 12, 96.

Lanza, S. T., & Collins, L. M. (2008). A new SAS procedure for latent transition analysis: Transitions in dating and sexual risk behavior. Developmental Psychology, 44(2), 446–456.

Lee, T., & Shi, D. (2021). A comparison of full information maximum likelihood and multiple imputation in structural equation modeling with missing data. Psychological Methods, 26(4), 466–485.

Little, R. J. A., & Rubin, D. B. (2002). Statistical analysis with missing data (2nd ed.). John Wiley & Sons.

Moore, E. W. G., Quartiroli, A., & Little, T. D. (2025). Introduction to the best practice recommendations for longitudinal latent transition analysis. International Journal of Psychology, 60(2), e70021.

Okpara, C., Edokwe, C., Ioannidis, G., Papaioannou, A., Adachi, J. D., & Thabane, L. (2022). The reporting and handling of missing data in longitudinal studies of older adults is suboptimal: A methodological survey of geriatric journals. BMC Medical Research Methodology, 22(1), 122.

Powney, M., Williamson, P., Kirkham, J., & Kolamunnage-Dona, R. (2014). A review of the handling of missing longitudinal outcome data in clinical trials. Trials, 15, 237.

Rubin, D. B. (1987). Multiple imputation for nonresponse in surveys. John Wiley & Sons.

Ryoo, J. H., Wang, C., Swearer, S. M., Hull, M., & Shi, D. (2018). Longitudinal model building using latent transition analysis: An example using school bullying data. Frontiers in Psychology, 9, 675.

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. Psychological Methods, 7(2), 147–177.

Scorza, P., Masyn, K. E., Salomon, J. A., & Betancourt, T. S. (2015). A latent transition analysis for the assessment of structured diagnostic interviews. Psychological Assessment, 27(3), 975–984.

Sorgente, A., Lanz, M., Serido, J., Tagliabue, S., & Shim, S. (2019). Latent transition analysis: Guidelines and an application to emerging adults' social development. Testing, Psychometrics, Methodology in Applied Psychology, 26(1), 39–72.

Sterba, S. K. (2016). A latent transition analysis model for latent-state-dependent nonignorable missingness. Psychometrika, 81(2), 506–534.